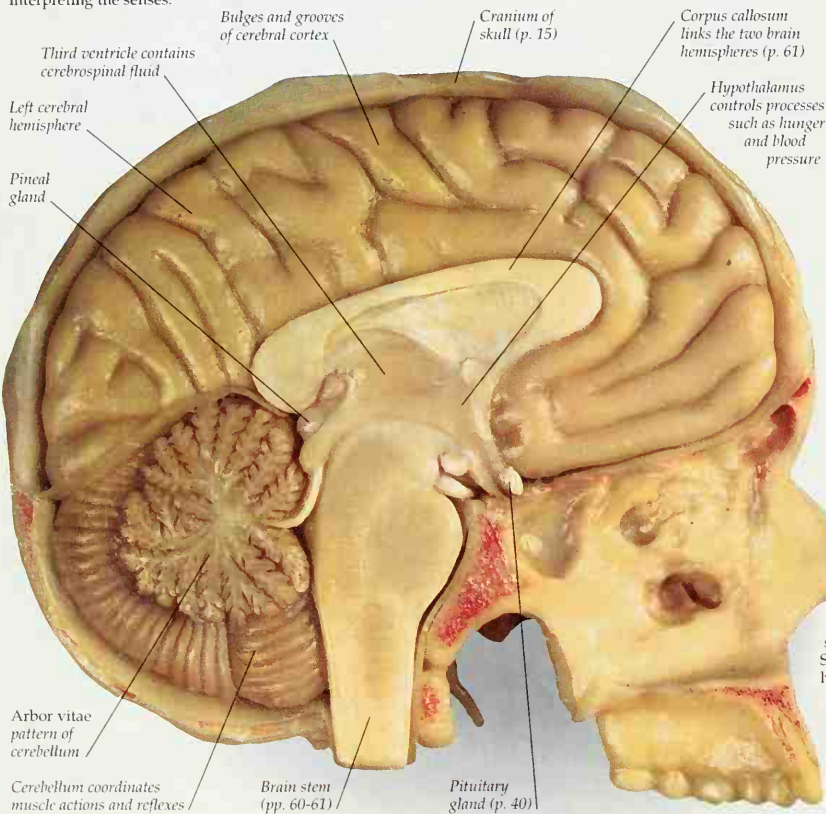


Inside the brain

THIS BOOK BEGAN BY COMPARING THE HUMAN BODY with its fellow mammals, and finding it similar – except for the brain. Recent scientific research seems to confirm that in this organ reside the secrets of consciousness, thoughts, reasoning, intelligence, memory, language, and other aspects of our unique “humanness.” Influential French philosopher and mathematician René Descartes (1596-1650) proposed a theory of the universe as a giant clockwork-type machine of minute particles of “matter” that moves according to mechanical principles. He said the human being, however, was a combination of two kinds of substances – body (matter) and mind, a thinking substance. The body existed and could be measured. The mind was different. It thought, sensed, and willed. The link between the two is the pineal gland, deep in the brain. Through the pineal, said Descartes, the mind influences parts of the brain. The effects travel through the fluid ventricle chambers and along the nerves, to produce muscle movements. Since Descartes’ time, the relationship of body and mind has been an endless source of discussion and fascination for philosophers and scientists alike.



LIQUID INTELLIGENCE? Galen correctly said that the brain's abilities could be attributed to the various regions of its solid parts. For centuries mainstream opinion had placed more importance on the pale yellow cerebrospinal fluid in the brain's ventricles, or inner chambers. This diagram from the 17th century links each chamber, or “cell,” with a mental quality such as imagination, or interpreting the senses.



A MAZE OF NERVE SIGNALS

Brain tissue is made up of billions of neurons, each with thousands of synaptic connections (p. 59), and support cells, providing mechanical and metabolic servicing for them.

This micrograph shows five star-shaped astrocytes, support cells in the brain's “gray matter.” A recent discovery shows that these cells may play a role in changing neuron connections according to outside stimulations, which could be the basis of memory.

Nose bone of skull (p. 15)

LOOKING INSIDE THE BRAIN

In this model the head is cut down the center, between the eyes, showing the central slice of the brain. Since the inner side of the cerebral hemisphere curves down to the midline, it has not been cut; the corpus callosum has. This bridge of over 100 million nerve fibers links the two sides of the brain (p. 61). The cerebellum shows its folded structure, which produces a branching pattern known as the *arbor vitae*, or “tree of life.”